



July 2024

**climate risk in vulnerable mountain
regions of the world**

Day 4 (2 July)

Jakob Steiner

Course content

28 June (1400 – 1700)

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

29 June (1000 – 1800)

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

Science to policy

30 June (1000 – 1800)

Serious game

2 July (1600 – 1800)

Risk mapping and DRR – self evaluation - recap

Course content

2 July (1600 – 1800)

16:00 – 17:20	Risk mapping and DRR
17:20 – 17:35	Course Recap and remaining questions
17:35 - 18:00	Self Evaluation

Disaster Risk Reduction

Disaster risk reduction (DRR) is defined by United Nations Office for Disaster Risk Reduction (UNDRR) as those actions which aim to "prevent new and reducing existing disaster risk and managing residual risk, all of which contribute to strengthening resilience and therefore to the achievement of sustainable development".



Disaster Risk Reduction – Sendai Framework

Chart of the Sendai Framework for Disaster Risk Reduction 2015-2030

Scope and purpose

The present framework will apply to the risk of small-scale and large-scale, frequent and infrequent, sudden and slow-onset disasters, caused by natural or manmade hazards as well as related environmental, technological and biological hazards and risks.
It aims to guide the multi-hazard management of disaster risk in development at all levels as well as within and across all sectors

Expected outcome

The substantial reduction of disaster risk and losses in lives, livelihoods and health and in the economic, physical, social, cultural and environmental assets of persons, businesses, communities and countries

Goal

Prevent new and reduce existing disaster risk through the implementation of integrated and inclusive economic, structural, legal, social, health, cultural, educational, environmental, technological, political and institutional measures that prevent and reduce hazard exposure and vulnerability to disaster, increase preparedness for response and recovery, and thus strengthen resilience

Targets

Substantially reduce global disaster mortality by 2030, aiming to lower average per 100,000 global mortality between 2020-2030 compared to 2005-2015

Substantially reduce the number of affected people globally by 2030, aiming to lower the average global figure per 100,000 between 2020-2030 compared to 2005-2015

Reduce direct disaster economic loss in relation to global gross domestic product (GDP) by 2030

Substantially reduce disaster damage to critical infrastructure and disruption of basic services, among them health and educational facilities, including through developing their resilience by 2030

Substantially increase the number of countries with national and local disaster risk reduction strategies by 2020

Substantially enhance international cooperation to developing countries through adequate and sustainable support to complement their national actions for implementation of this framework by 2030

Substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information and assessments to people by 2030

Priorities for Action

There is a need for focused action within and across sectors by States at local, national, regional and global levels in the following four priority areas.

Priority 1 Understanding disaster risk	Priority 2 Strengthening disaster risk governance to manage disaster risk	Priority 3 Investing in disaster risk reduction for resilience	Priority 4 Enhancing disaster preparedness for effective response, and to «Build Back Better» in recovery, rehabilitation and reconstruction
Disaster risk management needs to be based on an understanding of disaster risk in all its dimensions of vulnerability, capacity, exposure of persons and assets, hazard characteristics and the environment	Disaster risk governance at the national, regional and global levels is vital to the management of disaster risk reduction in all sectors and ensuring the coherence of national and local frameworks of laws, regulations and public policies that, by defining roles and responsibilities, guide, encourage and incentivize the public and private sectors to take action and address disaster risk	Public and private investment in disaster risk prevention and reduction through structural and non-structural measures are essential to enhance the economic, social, health and cultural resilience of persons, communities, countries and their assets, as well as the environment. These can be drivers of innovation, growth and job creation. Such measures are cost-effective and instrumental to save lives, prevent and reduce losses and ensure effective recovery and rehabilitation	Experience indicates that disaster preparedness needs to be strengthened for more effective response and ensure capacities are in place for effective recovery. Disasters have also demonstrated that the recovery, rehabilitation and reconstruction phase, which needs to be prepared ahead of the disaster, is an opportunity to «Build Back Better» through integrating disaster risk reduction measures. Women and persons with disabilities should publicly lead and promote gender-equitable and universally accessible approaches during the response and reconstruction phases

Guiding Principles

Primary responsibility of States to prevent and reduce disaster risk, including through cooperation	Shared responsibility between central Government and national authorities, sectors and stakeholders as appropriate to national circumstances	Protection of persons and their assets while promoting and protecting all human rights including the right to development	Engagement from all of society	Full engagement of all State institutions of an executive and legislative nature at national and local levels	Empowerment of local authorities and communities through resources, incentives and decision-making responsibilities as appropriate	Decision-making to be inclusive and risk-informed while using a multi-hazard approach
Coherence of disaster risk reduction and sustainable development policies, plans, practices and mechanisms, across different sectors	Accounting of local and specific characteristics of disaster risks when determining measures to reduce risk	Addressing underlying risk factors cost-effectively through investment versus relying primarily on post-disaster response and recovery	«Build Back Better» for preventing the creation of, and reducing existing, disaster risk	The quality of global partnership and international cooperation to be effective, meaningful and strong	Support from developed countries and partners to developing countries to be tailored according to needs and priorities as identified by them	

Target A: Disaster-related deaths and missing persons (excluding COVID-19)

- Average disaster-related mortality per 100,000 population declined from 1.60 in 2005-2015 to 1.15 in 2013-2022 (a decline of 28%).
- During 2015-2022, the average annual disaster-related mortality stood at 41,789 persons in absolute terms.

Target B: Disaster-affected population

- Number of disaster-affected people increased from 1,092 per 100,000 population during 2005-2015 to 2,034 during 2013-2022 (an increase of 86%).
- On an average, more than 130 million people were affected by disasters each year globally during 2015-2022.

Target C: Disaster-related economic losses

- Direct economic loss averages 0.37% of global GDP of reporting countries during 2015-2022.
- Provisional data shows that in 2022, direct economic loss accounts for 0.83% of global GDP.

Target D: Disaster damage to critical infrastructure and disruption of basic services

- From 2015 to 2022, over 800,000 critical infrastructure units and facilities, including schools and hospitals, were damaged or destroyed due to disasters.
- In the same period, more than 12 million educational, health, and other basic services were disrupted by disasters.

Target E: National and local disaster risk reduction (DRR) strategies

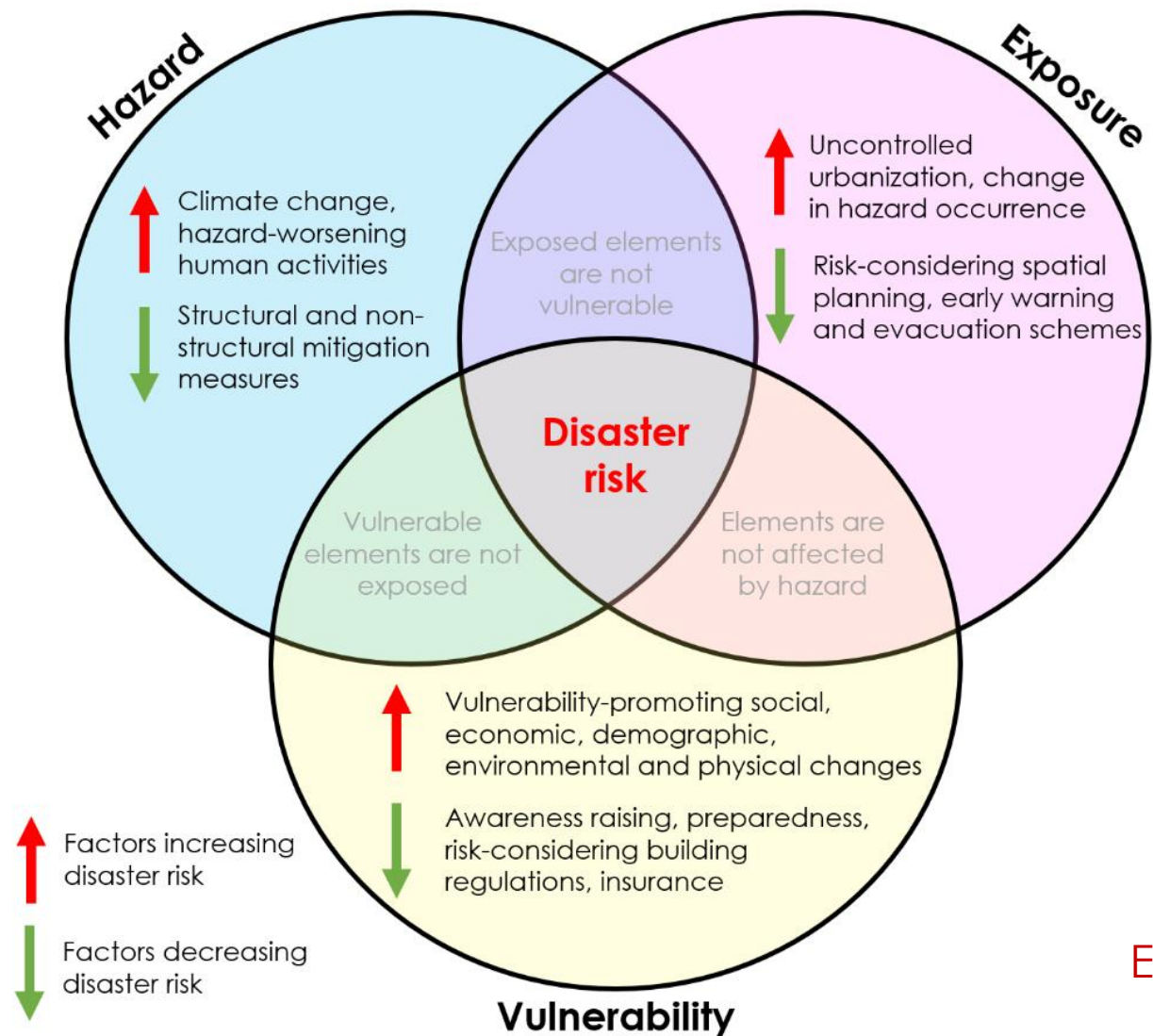
- 126 countries have reported having national disaster risk reduction strategies, compared to 55 in 2015 (two-third of countries in the world and an increase of 130% since the adoption of Sendai Framework).
- 102 countries have reported having local disaster risk reduction strategies. Average proportion of local governments in such countries is 72%.

Target F: International cooperation for disaster risk reduction

- International cooperation for developing countries remains limited, as developing countries reported only US\$ 304 million of official development assistance (ODA) plus other official flows for supporting disaster risk reduction actions in 2021.
- From 2005 to 2022, 1,742 programmes and initiatives on science, technology and innovation transfer in disaster risk reduction were reported by 39 countries.
- In the same period, 33,441 instances of capacity development on disaster risk reduction were reported by 42 countries.

Target G: Multi-hazard early warning systems (MHEWS)

- 101 countries reported having MHEWS, compared to 47 countries in 2015 (an increase of 114%).
- Two third (66%) of the countries reported having MHEWS have self-assessed their MHEWS with "limited to moderate achievement."



Emmer, Steiner and Kalvoda (2024)

Risk management – disaster risk reduction

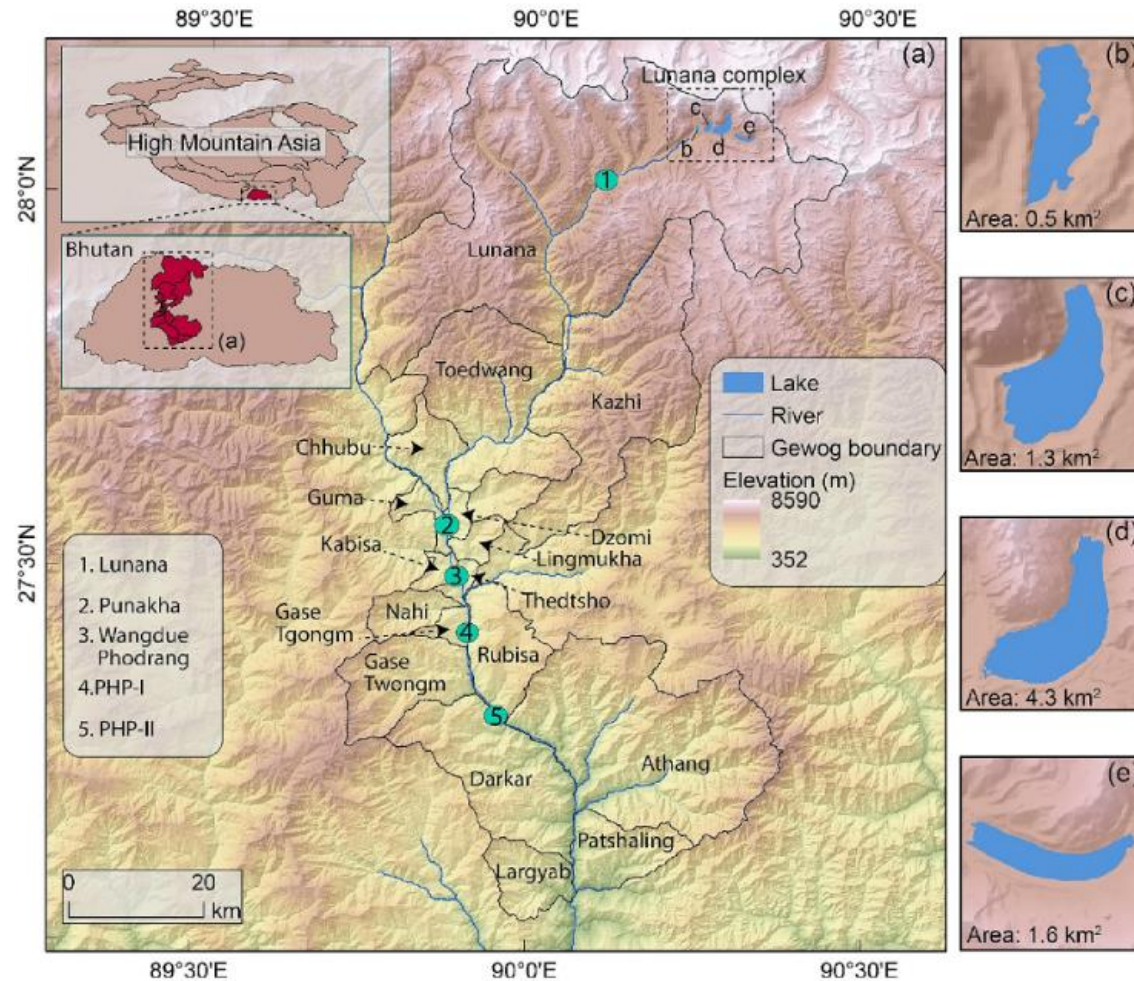


Fig. 1. Lunana glacier complex and downstream settlements. Map (a) shows settlements downstream of the Lunana glacier complex along the Phochu and Punatsangchu basins. Downstream settlements are presented following the subdistrict block administrative boundary. Panels (b–e) show four glacial lakes, which we studied here: Bechung Tsho, Raphstreng Tsho, Lugge Tsho, and Thorthormi Tsho, respectively. Background image: ALOS-PALSAR DEM.

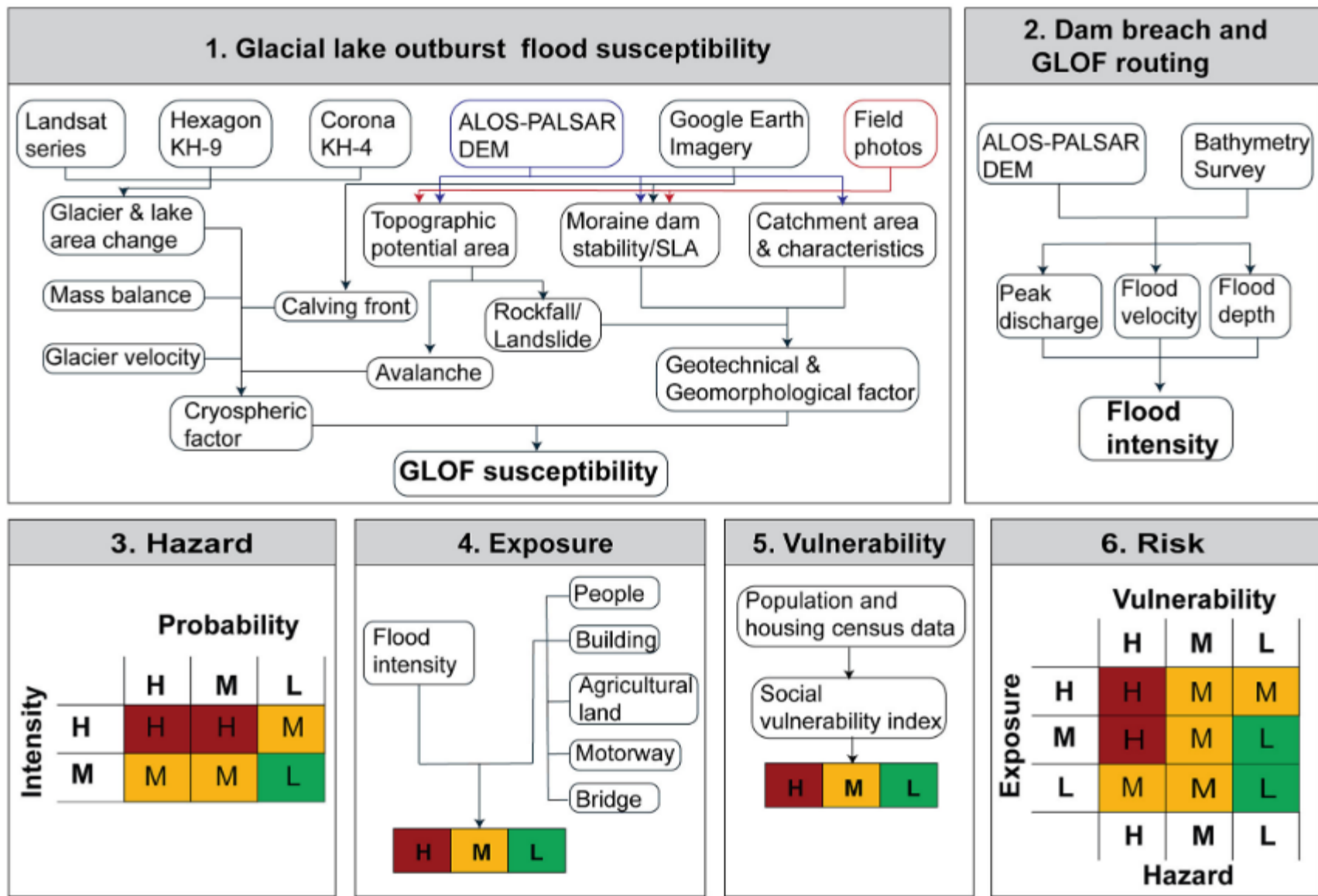


Fig. 2. Methodological flow chart adopted for GLOF susceptibility, downstream hazard, exposure, vulnerability, and risk assessments for the potentially dangerous glacial lakes in the Lunana glacier complex of the Bhutan Himalaya: downstream GLOF hazard, vulnerability, exposure, and risk are categorized into three classes; high (H), medium (M) and low (L). For the hazard computation, the GLOF intensity used here is translated from the modelled flood depth: high (more than 1 m) and medium (less than 1 m) (after [GAPHAZ, 2017](#)).

Hazards

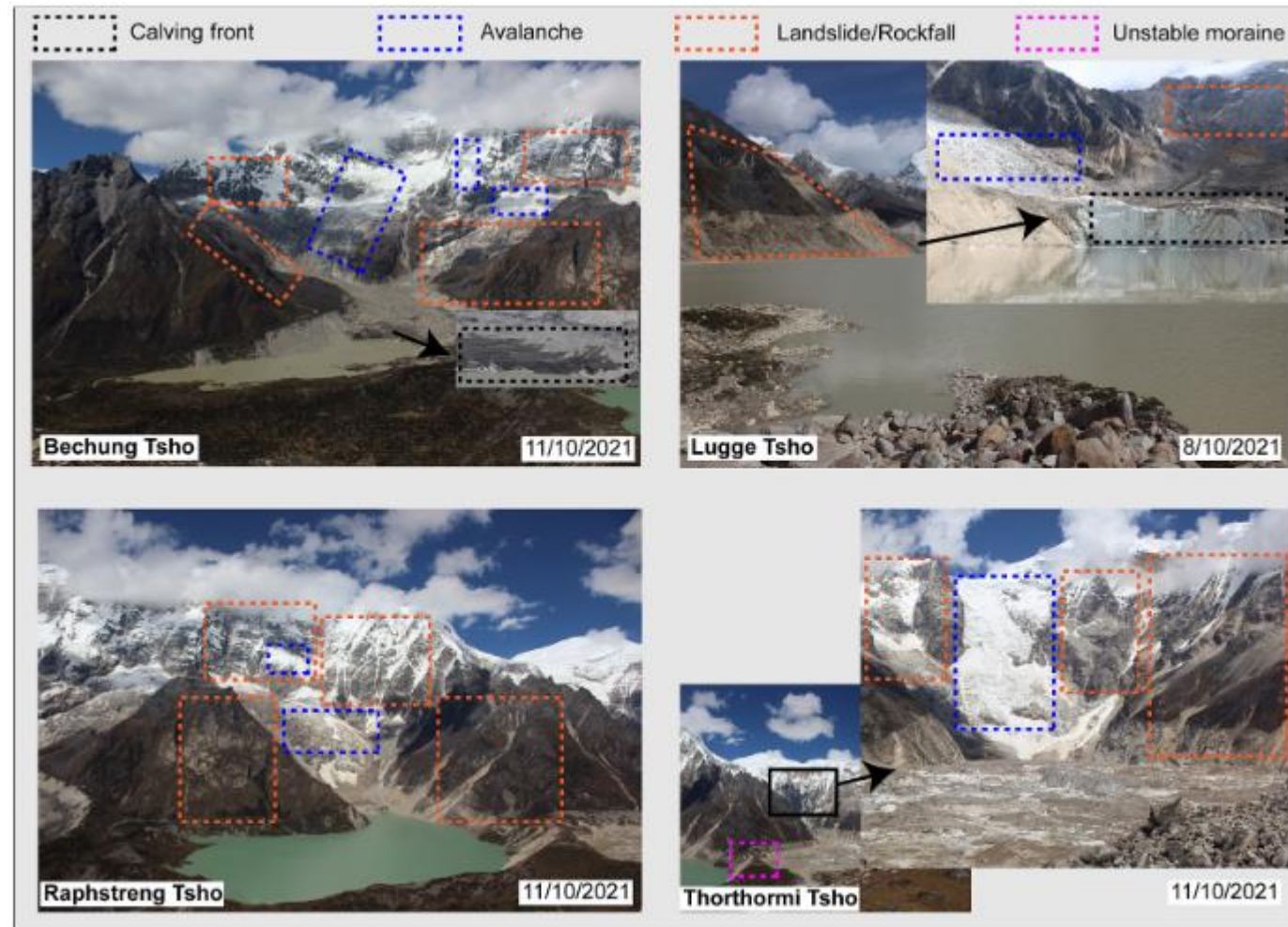
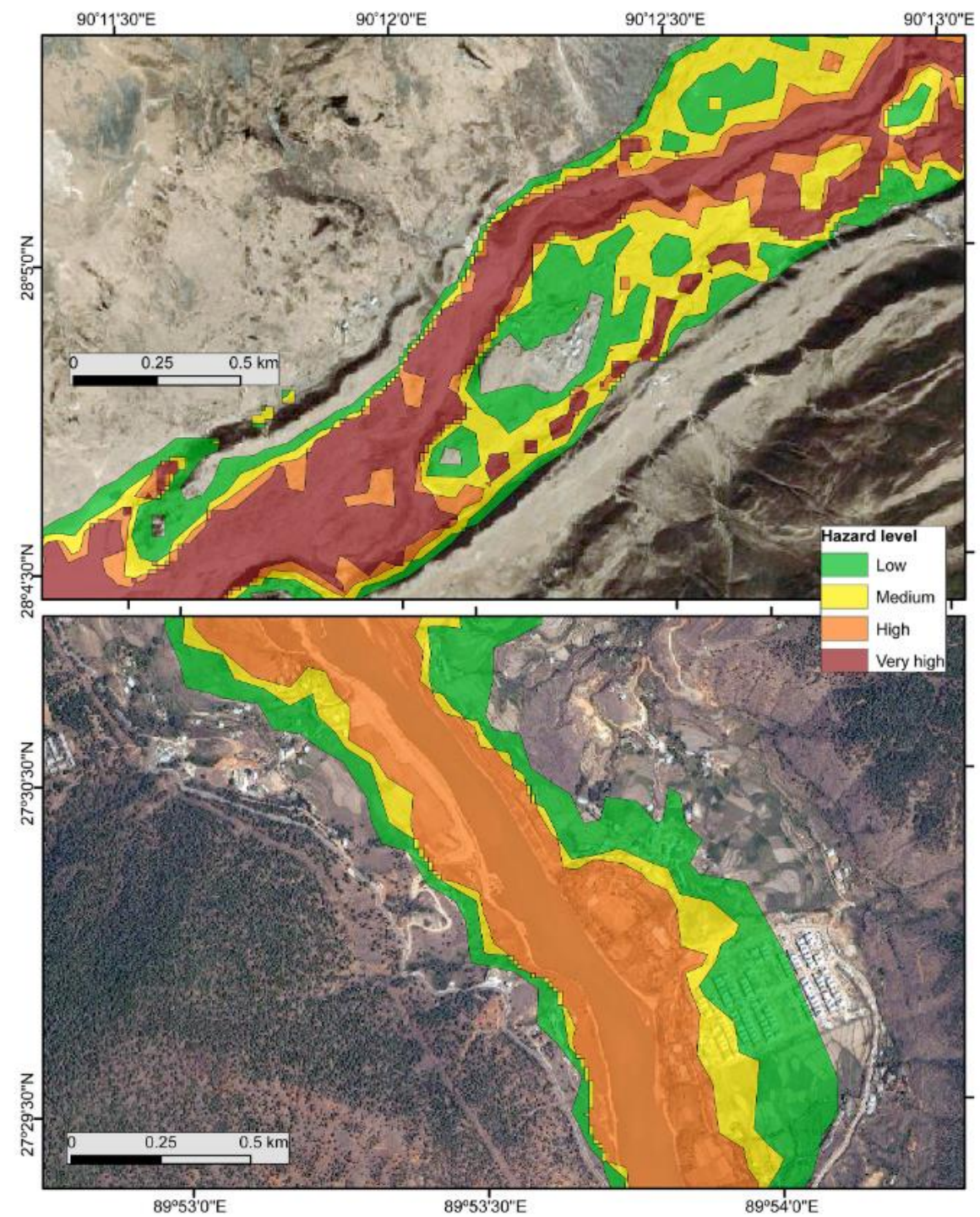


Fig. 3. Field photographs show the surrounding conditions that make lakes susceptible to GLOF, including calving front, avalanche, landslide/rockfall, and unstable moraine dam. Photo courtesy: NCHM, Bhutan.

Exposure



Vulnerability

Table 2

List of indicators and corresponding represented components considered for calculating the social vulnerability index for the downstream settlements. The indicators are extracted from the Bhutan Population and Housing Census 2017 data (NSB, 2018), following the convention used by Allen et al. (2016) and Cutter et al. (2003). The absolute and normalized values for each indicator are also provided in supplementary Table S2.

No.	Social vulnerability indicator	Component represented
1	Old age dependence ratio	Sensitivity, capacity to prepare, respond and recover
2	Child dependence ratio	Sensitivity, capacity to prepare, respond and recover
3	Disability prevalence rate	Sensitivity, capacity to prepare, respond and recover
4	Female population	Sensitivity, capacity to prepare, respond and recover
5	Household without high school education attainment (adequate literacy)	Capacity to prepare, respond and recover
6	Household with insufficiency in food	Sensitivity, capacity to prepare, respond and recover
7	Rented household	Capacity to recover
8	Household without a reliable source of water	Capacity to prepare, respond and recover
9	Household without an improved sanitation	Capacity to recover
10	Household with more than a 30-minute walk to the road point	Capacity to prepare, respond and recover
11	Household without a smartphone	Capacity to prepare and respond
12	Household living in the house without housing concrete wall	Capacity to recover
13	Household without a car	Capacity to prepare, respond and recover
14	Household without a TV	Capacity to prepare and respond
15	Household without electricity as a main source of energy	Capacity to prepare and respond

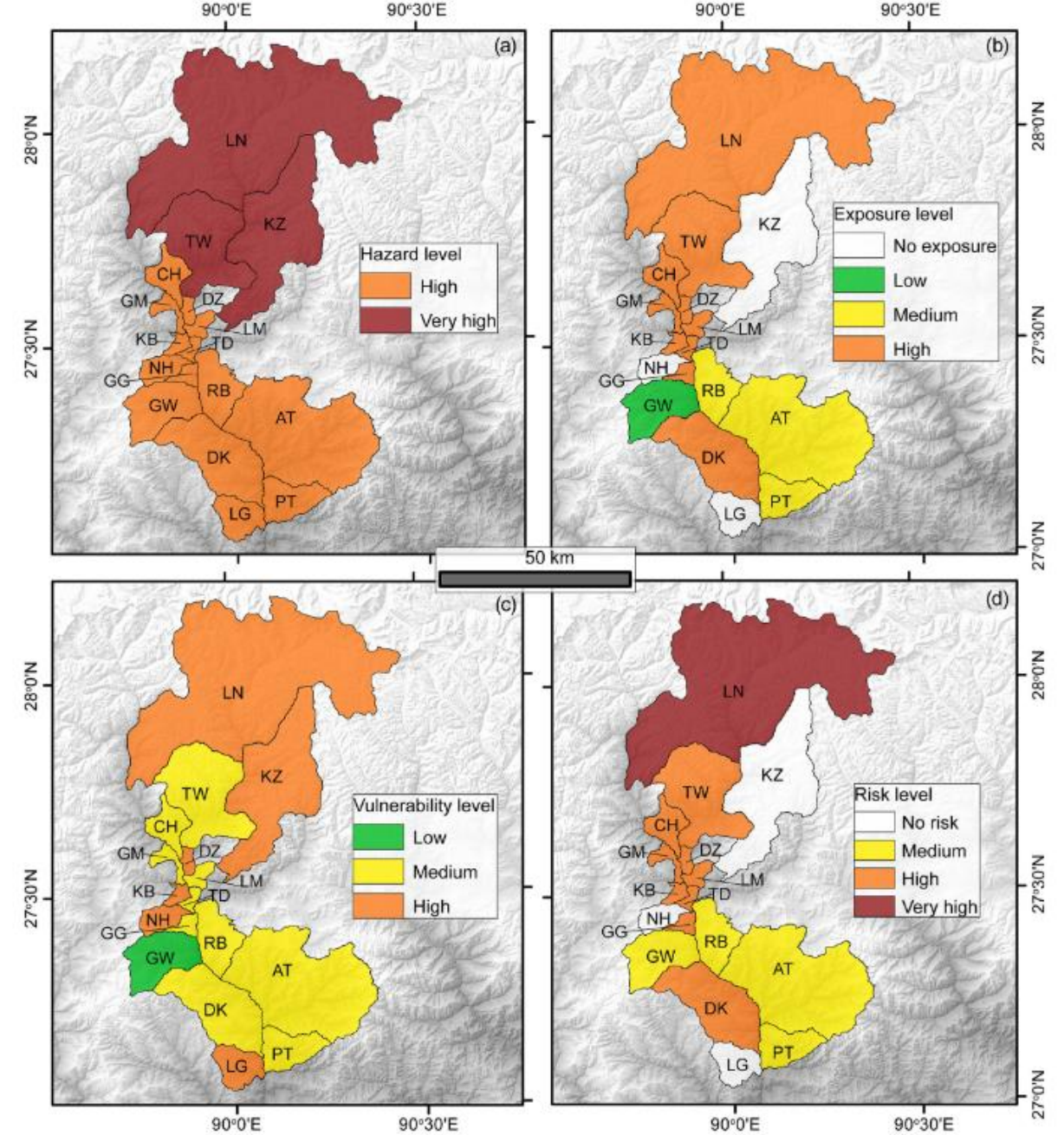


Fig. 9. The hazard (a), exposure (b), vulnerability (c), and risk (d) levels in 2020 of 17 subdistrict blocks downstream of the Lunana glacier complex under three different scenarios of GLOF magnitude (large, medium, and small) from each lake. The risk is calculated as a function of hazard, exposure, and vulnerability. The 17 subdistrict blocks are: Lunana (LN), Kazhi (KZ), Toedwang (TD), Chhubu (CH), Guma (GM), Dzomi (DZ), Lingmukha (LM), Kabisa (KB), Thedtsho (TD), Nahi (NH), Gase Tshogongm (GG), Gase Tshowongm (GW), Rubisa (RB), Darkar (DK), Athang (AT), Patshaling (PT), Larygab (LG). Background image: ALOS-PALSAR DEM.

RISK ASSESSMENT AND RISK MANAGEMENT PLAN

Risk Description	Risk Assessment	Mitigation Measures or Risk Management Plan
Construction and completion	Medium	The completion risk of any hydropower projects falls mainly on geological conditions and their relevant time and cost overruns. Many geological investigations and drilling studies undertaken for more than 3.5 years during project preparation found 80% of the rock tested to be in good order. The detailed project report prepared by a qualified engineering firm and the PPTA conducted due diligence studies including extensive field and laboratory investigations, such as hydrological, geological, geotechnical and geophysical, sedimentation, mechanical, and electrical requirements. Uncertainties of geological conditions remain as residual risks of construction delays and cost overruns, which could be affected by price increases in labor and materials. To address these risks, contingencies total \$32 million or 22% of the base cost. Contractual mitigations are included in engineering, procurement, and construction contracts with liquidity damage clauses on a milestone basis. DGPC has adequate experience in procuring and managing contractors for construction of hydropower projects, demonstrated with Dagachhu hydropower plant implementation. Experienced staff of Dagachhu will support project construction. Project site accessibility is very good, i.e., along the main east–west national highway close to access roads and transmission connections.
Operations	Low	The plant's run-of-river hydropower operations are straightforward. DGPC has operated five run-of-river hydropower plants (HPPs) with capacity of 60 MW, 64 MW, 126 MW, 336 MW, and 1,020 MW. THyE will be staffed with experienced staff from DGPC and its subsidiaries—management and staff with adequate experience in operating HPPs. DGPC plans to select a reputed private joint venture partner with extensive experience in similar project development. The project's powerhouse will release water discharge to the Mangdechhu HPP reservoir (720 MW, run-of-river), expected to be available in 2017. The reservoir, under construction, is large enough to absorb the project's water release (confirmed by the Mangdechhu HPP's dam safety assessment under the PPTA due diligence studies). Project operation can begin regardless of completion of Mangdechhu HPP construction, as the existing Mangdechhu's diversion tunnel could easily absorb the project's water discharge.
Offtake	Low	A long-term 25-year PPA was agreed with PTC, the leading power trading company in India. DGPC has already exported almost 75% of its generation through other PPAs with PTC. The project's PPA is a take-or-pay contract in fixed volume terms for 80% of salable power; the volume will be less than 1% of the entire PTC trading power, which PTC can easily absorb. Given the prolonged power shortage in India, all of the project's power will be readily absorbed. If PTC fails to pay as per the PPA, the right for third party sale will be triggered, but PTC's financial

Risk Description	Risk Assessment	Mitigation Measures or Risk Management Plan
Transmission	Low	was tested and the result showed that THyE would keep the breakeven point even if it encountered 15% revenue reduction from the base-case revenues for 25 years. This scenario is considered unlikely. The 132 kV transmission line works are contracted with BPC for project's full annual operations in 2020. The nearest substation under construction is on track to be available in 2017 as part of the ongoing Mangdechhu HPP. Once the project's transmission line (18.6 km) is connected to the substation, the project's power can alternatively be exported through existing grid connections to India. The project's power will be mostly evacuated to India through the Mangdechhu–Golmud–Jigmeiling–Alipurduar transmission line corridor (400 kV) under construction for Mangdechhu operations scheduled in 2017.
Hydrological	Low	The hydrological risk primarily comprises drought risks, which could significantly affect hydropower generation. The project's design energy is based on 22-year hydrological data; the risk of any deviation is considered minimal. The financial analysis is based on the design energy of 90% dependability (P-90 generation and plant availability of 95%) which is quite conservative. This means that more water is likely to be available for 90% of the operating time than designed on the hydrological data. The assessment of potential future hydrological impacts, using various climate models found the project to be hydrologically sound. The projects that annual water flow in the Nikachhu watershed would increase by an average by 1%–3% for non-monsoon months and 5%–8% for monsoon months to 2041–2060.
Other climate change (disaster)	Low	The climate change modeling analysis was extended to risks other than potential hydrological issues, including risks of possible floods, earthquakes, landslides, and glacial lake outbursts. The engineering design and environmental management plan are adaptive to extreme severe conditions created by such events. The analysis concludes that overall climate change is unlikely to impose negative effects on the project and its operation. Details of the assessment are in the climate change impact assessment. Effective adaptation measures will be adopted for the project's structures, facilities, and adjoining areas to address such potential disasters as floods, landslides, and earthquakes. The dam safety design was reviewed and continues to be monitored by DGPC's dam safety committee and a panel of experts. The project will include an early warning system and awareness programs for public safety in spite of no human settlement along the river downstream to the confluence with the main river of the Mangdechhu.
Funding and debt	Low	The cofinancing loan from Indian commercial banks has been approved in principal. The cofinancing framework was agreed and its signing

Currency	Medium	Since the project is earned in Indian rupees from power exports, foreign debt from the OCR loan will have a currency exchange rate fluctuation
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Risk Description	Risk Assessment	Mitigation Measures or Risk Management Plan
		risk if the Indian rupee depreciates. The Indian commercial bank cofinancing in local terms will reduce the overall debt service risk. The PPA stipulates that tariff adjustments in dollar terms may be introduced sometime in the future if both parties agree. Given the currency risk assessment, THyE will fully meet the debt service requirements even under an extreme scenario of continued annual 7% depreciation of Indian rupees over the debt repayment period. To mitigate the residual risks arising from the foreign debt services, THyE will adopt a foreign exchange fluctuation reserve fund. THyE and DGPC will also explore periodic currency hedging contracts once this becomes a feasible option in the financial market as covenanted.
Safeguards	Low	All safeguard documents are in accordance with ADB's Safeguard Policy Statement (2009). The compensation relating to resettlement will be fully paid in a satisfactory manner to the affected people. The current mitigation measures in the environmental management plan are appropriate and adequate to contain environmental impacts during project construction and operation, and environmental monitoring will be carried out during the project. To ensure minimum ecological flows, particularly in the lean seasons, the recommended level of 10% will continue to be monitored and finalized during implementation to ensure the habitat requirements combined with integrated water resources management, including relevant plant operations in the river basin. The ADB financing and loan agreements will ensure compliance with safeguard measures and implementation requirements, and an attached TA will support the safeguard activities. Extensive consultations and

Financial management	Low	monitor and audit implementation of safeguard measures. DGPC has adopted comprehensive control systems covering accounting and financial management, procurement management, internal audit, and environmental safeguards, along with enterprise resource planning. An external independent auditor reported that DGPC has adequate internal control systems, competitive bidding, and strict control to carry out their operations in accordance with the PPA. THyE will follow the same policies and procedures.
Procurement	Low	Increased transparency, efficiency, and compliance with ADB requirements for major construction contracts. DGPC will conduct regular site inspections and ensure the timeliness of progress on contracts where applicable.
Corruption	Low	Increased transparency and accountability. DGPC requirements and regular stakeholder consultations and agreements ensure that ADB's Anticorruption measures are conformed with. ADB will monitor and audit expenditure related information on the website, and ensure transparency in expenditure by ADB.
Overall	Low	

ADB = Asian Development Bank, BPC = Bhutan Power Corporation, DGPC = Druk Green Power Corporation, KV = kilovolt, MW = megawatt, OCR = ordinary capital resources, PPA = power purchase agreement, PTA = preparatory technical assistance, PTC = Power Trading Corporation of India, THyE = Tamsang Hydro Energy, Tamsang Hydro Electric Project.

Source: Asian Development Bank.

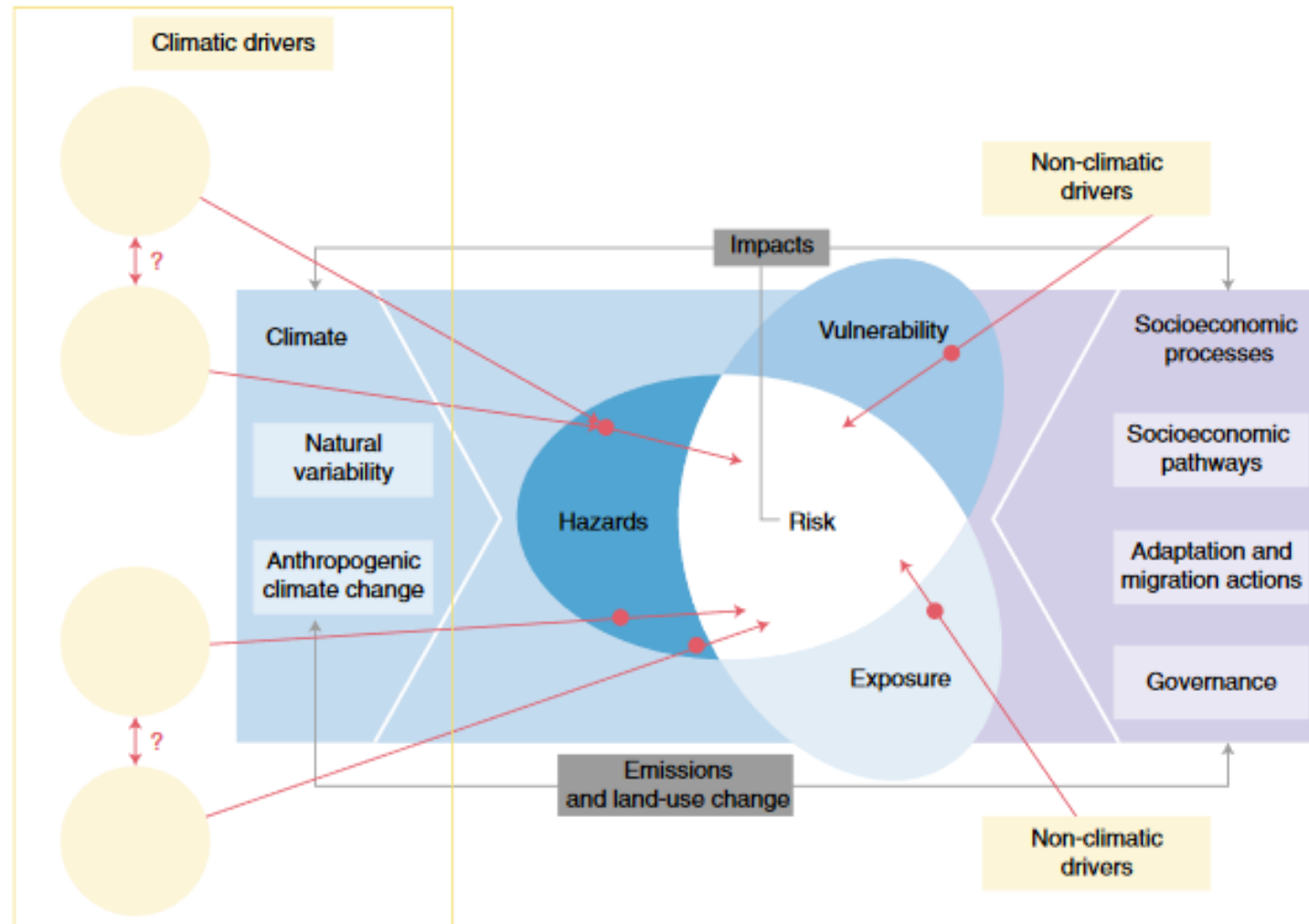
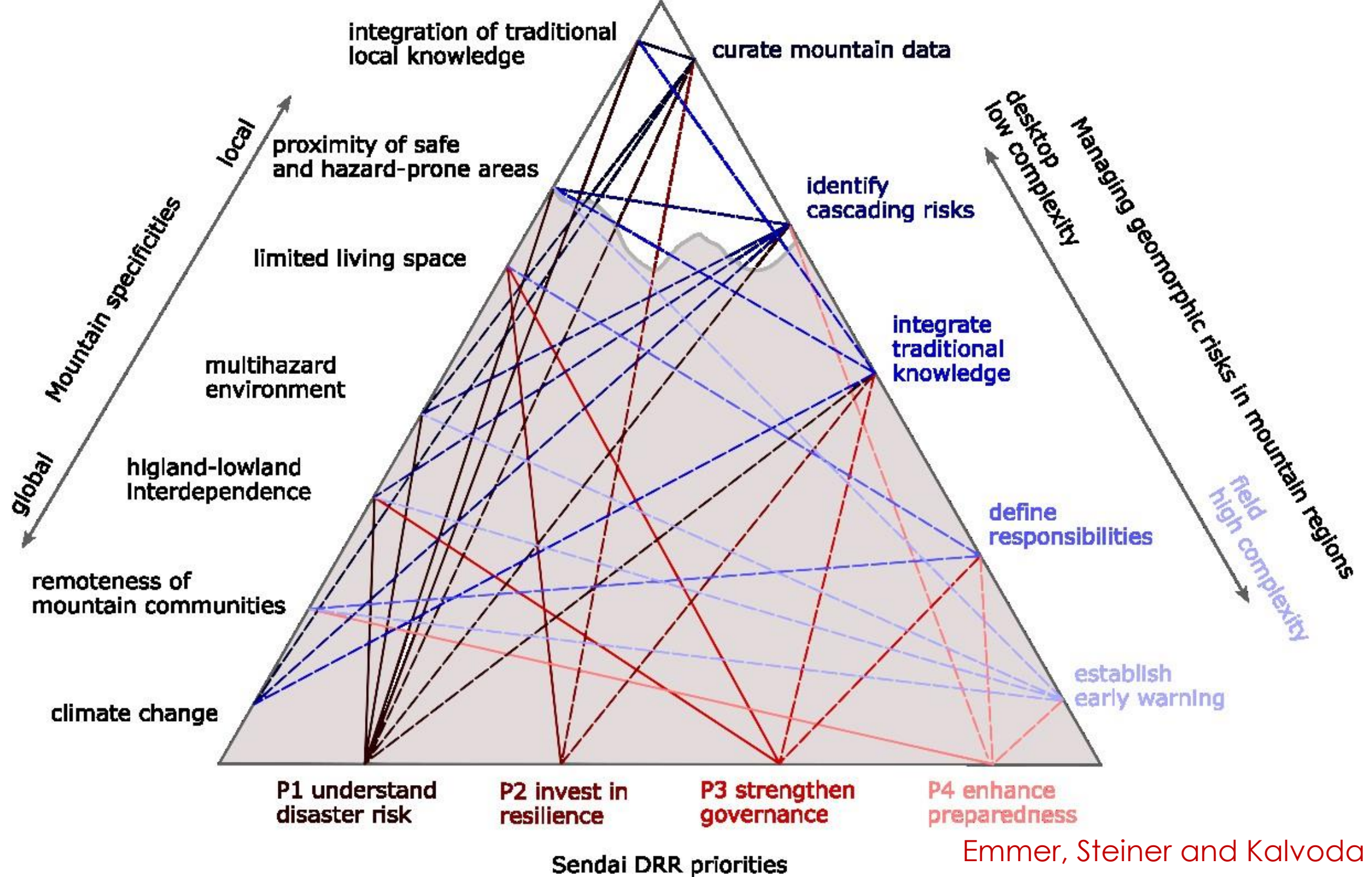


Fig. 1 | Extended risk framework. Multiple climatic drivers cause one or multiple hazards leading to societal and environmental risk. The climate drivers (which may vary from local-scale weather to large-scale climate modes, represented by yellow circles) and/or hazards may be mutually dependent. Non-climatic drivers related to vulnerability and exposure may also contribute to risk. Background risk figure adapted from ref. ⁹⁶, IPCC.



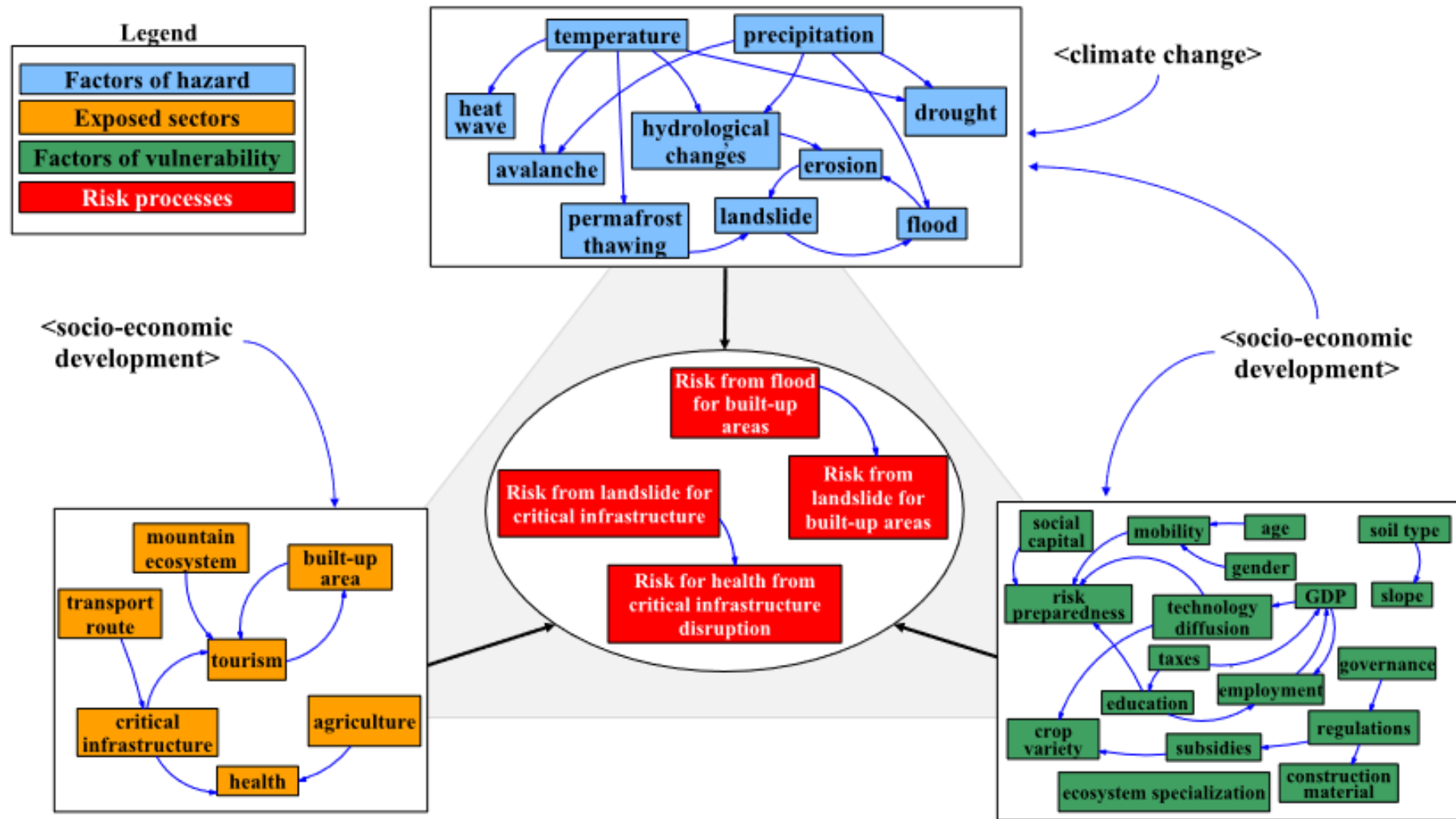


Fig. 2. Conceptual framework for mountain regions expanding the risk components into multi-hazard, exposure, multi-vulnerability and multi-risk.







Soneri Bank

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کی تمام براچرز سے وصول کی جاسکتی ہیں















LONG LIVE
PAK-CHINA FRIENDSHIP



中巴友谊万岁

PAK FC P52
OKZ



آئے والے تمام مہمانوں کو خوش آمدید
Pak-FC

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We're done ... time for a look back

Your final thoughts on risks in mountain regions?

Your questions?

Course self evaluation (Google Form, Moodle)

University Tübingen course evaluation

Literature

Sonam Rinzin, Guoqing Zhang, Ashim Sattar, Sonam Wangchuk, Simon K. Allen, Stuart Dunning, Menger Peng, GLOF hazard, exposure, vulnerability, and risk assessment of potentially dangerous glacial lakes in the Bhutan Himalaya, *Journal of Hydrology*, Volume 619, 2023

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